

Remittance Corridor Analyzer

PoC ID 13 · Payment Rail · Geo/Temporal Archetype

AUDITOR	Real Rails AI Architect
DATE	May 09, 2026
VERSION	v4.0 — Feature Expansion (5 New Features)
STATUS	AUDIT COMPLETE

FIELD	VALUE
Title	Remittance Corridor Analyzer
Rail	Payments
Topic	Cost, speed, and route quality for remittance corridors
Best Format	Episode + LinkedIn
Data Sources	World Bank Data; ECB Data Portal
Libraries	React; Next.js; TypeScript; Tailwind CSS; shadcn/ui; Leaflet; React Leaflet; D3.js; TanStack Table; FastAPI; Pandas; GeoPandas
Mock Data	Yes — synthetic corridor transaction samples for unit economics
Difficulty	Medium
Why It Matters	Strong social + economic framing for payments rails that people already use.

Acting as Senior UX Architect, auditing the Remittance Corridor Analyzer codebase (backend/main.py, page.tsx, CorridorMap.tsx, api.ts, Sidebar.tsx) against the Real Rails DNA and Excel Row 13 requirements.

#	AUDIT CATEGORY	CHECK	FINDING	VERDICT
A1	Requirement Match Archetype	Visual archetype matches Excel's intent (Geo map + Temporal timeline = Episode + LinkedIn)	CorridorMap.tsx renders a Leaflet dark-tile world map with polyline arcs per corridor. TimelineChart.tsx adds send/receive volume trend. Archetype is correctly Geo + Temporal as mandated for an Episode-format PoC.	✓ PASS

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A2	Requirement Match Project Identity	Title, Rail, and PoC ID match Row 13	page.tsx header: 'Remittance Corridor Analyzer · PAYMENT RAIL · ID 13'. backend/main.py title: 'Real Rails – Remittance Corridor Analyzer v3.0.0'. Perfect match.	✓ PASS
A3	Requirement Match Required Features	All 5 required features present: corridor map, cost slider, send/receive timeline, informal vs formal comparison, provider benchmarking	Corridor map: CorridorMap.tsx ✓. Cost slider: amount state + slider in page.tsx ✓. Send/receive timeline: TimelineChart.tsx ✓. Informal vs formal: channel filter ('all'/'formal'/'informal') ✓. Provider benchmarking: ProviderChart.tsx ✓. All 5 features implemented.	✓ PASS
A4	Requirement Match Mandatory Panels	'Why this matters' panel present in sidebar	Sidebar.tsx Section B contains 'WHY THIS MATTERS' with contextual remittance copy explaining cost burden on migrant workers and the G20 5% target. Present and populated.	✓ PASS
A5	Requirement Match Mandatory Panels	'Who controls the rail' panel present in sidebar	Sidebar.tsx Section C 'WHO CONTROLS THE RAIL' lists World Bank, SWIFT/Correspondent Banks, MTOs (Western Union, Wise, Remitly), and Central Banks. Matches ECB + World Bank data sources.	✓ PASS
A6	Requirement Match Filters & Tooltips	Filters, tooltips, and downloadable sample data present	Channel filter (all/formal/informal) and amount slider in page.tsx ✓. CorridorMap tooltips with cost%, volume, speed ✓. Download CSV endpoint at /api/download ✓.	✓ PASS
B1	DNA Check Background Colour	Background is strictly #030712 (Obsidian Black — mandatory)	globals.css: --rr-bg: #030712. page.tsx root div: bg-rr-bg. layout.tsx: bg-rr-bg on body. DNA constraint satisfied.	✓ PASS
B2	DNA Check 70/30 Split	Main stage = 70%, Intelligence sidebar = 30%	page.tsx: main stage uses style={{width:'70%'}} and sidebar uses style={{width:'30%'}} with flex-none class applied. Resolution-independent 70/30 split confirmed. Fix applied.	✓ PASS
B3	DNA Check Font Stack	Space Grotesk (primary) + IBM Plex Mono (labels/mono)	globals.css imports both fonts from Google Fonts: Space Grotesk and IBM Plex Mono. CSS variables --rr-font-sans and --rr-font-mono applied. DNA fonts confirmed.	✓ PASS
B4	DNA Check Colour Palette	Primary: #38BDF8 (cyan), Secondary: #818CF8 (indigo), Alert: #F87171, Surface: #0B1117	tailwind.config.js defines rr-cyan: #38BDF8, rr-indigo: #818CF8, rr-red: #F87171, rr-surface: #0B1117. All DNA colours correctly tokenised and applied throughout.	✓ PASS
B5	DNA Check Header Bar	Top bar with pulsing status dot, project name, archetype badge	page.tsx header: pulsing dot (pulse class), 'Remittance Corridor Analyzer', 'PAYMENT RAIL · ID 13' badge, data source badge (LIVE/MOCK indicator). Compliant.	✓ PASS

#	AUDIT CATEGORY	CHECK	FINDING	VERDICT
B6	DNA Check Dark Theme Integrity	No light-mode leakage, no white backgrounds on cards	All card backgrounds use rr-surface (#0B1117) or rr-bg (#030712). Border colour rr-border (#1F2937) consistently applied. No light-mode classes detected.	✓ PASS
C1	Data Mapping World Bank Live API	World Bank remittance cost data fetched as primary source	backend/main.py fetch_worldbank_remittance_costs() calls World Bank Remittance Prices API. Auto-fallback to mock_data.json on failure. Data source badge reflects live vs mock status. Live data path confirmed in code.	✓ PASS
C2	Data Mapping ECB API	ECB Data Portal integrated as secondary source	backend/main.py fetch_ecb_fx_rates() calls ECB data-api for EUR/USD and EUR/MXN exchange rates. Used in corridor cost calculations. Fallback to mock rates provided. ECB integration confirmed.	✓ PASS
C3	Data Mapping Mock Data Labelling	Synthetic data clearly labelled (no false live-data claims)	mock_data.json used for unit economics (tx samples). api.ts logs: '[RealRails] Backend offline — frontend mock active'. Data source badge in header shows MOCK vs LIVE. Protocol satisfied.	✓ PASS
C4	Data Mapping 70% Stage Accuracy	Corridor map in main stage accurately reflects data source findings	CorridorMap renders corridors from API /api/corridors response. Each corridor arc shows from/to coordinates, volume, avg_cost_pct, and speed_days from World Bank data. Selected corridor highlights in cyan with full metadata tooltip. Accurate mapping confirmed.	✓ PASS
C5	Data Mapping Sidebar Intelligence	30% sidebar populates with correct corridor metadata on map click	onSelect(id) in CorridorMap.tsx calls setSelectedId in page.tsx. Sidebar receives 'active' corridor prop and renders corridor-specific: volume, avg cost, speed, formal/informal split, and provider list. Handshake verified in code.	✓ PASS
C6	Data Mapping Provider Benchmarking	Provider chart shows real provider cost comparison per corridor	mock_data.json updated: 46 corridor-specific provider entries across 15 corridors. main.py /api/cost-analysis filters MOCK_PROVIDERS by corridor param. Sidebar.tsx passes corridor.id to getCostAnalysis(). api.ts mockProviders() filters by corridorId. Expanded from 7 to 15 corridors. Tested in browser — CONFIRMED.	✓ PASS
D1	Functional Data Fallback Chain	3-tier fallback: Live API → FastAPI mock → Frontend mock	api.ts safeFetch() catches HTTP errors and returns frontend MOCK data. backend/main.py wraps each live call in try/except returning MOCK_CORRIDORS. 3-tier chain correctly implemented. App never shows blank state.	✓ PASS
D2	Functional Amount Slider	Cost slider updates displayed send/receive amounts dynamically	page.tsx: amount state (default 200) wired to slider input. Sidebar receives amount prop and computes recipient_amount = amount * (1 - cost_pct/100) / fx_rate. Dynamic update confirmed.	✓ PASS

#	AUDIT CATEGORY	CHECK	FINDING	VERDICT
D3	Functional Channel Filter	Formal/Informal toggle filters timeline and stats correctly	channel state ('all'/'formal'/'informal') passed to TimelineChart and Sidebar. TimelineChart filters rows by channel prop. Volume and tx_count update accordingly. Filter logic confirmed in code.	✓ PASS
D4	Functional Leaflet Map Init	Map initialises once without 'already initialised' crash	CorridorMap.tsx Fix 1 & 4: checks container._leaflet_id before init, deletes stale ID on remount, returns cleanup that calls map.remove(). React Strict Mode double-invoke handled. Map init is safe.	✓ PASS
D5	Functional Export / Download	Downloadable sample data available	backend/main.py: GET /api/download returns StreamingResponse CSV of corridor data. Sidebar Section E contains download link pointing to API endpoint. Export feature implemented.	✓ PASS
D6	Functional Responsiveness	Dashboard functional at standard viewport (1280px+)	Tailwind CSS utility classes used throughout. Header uses flex/justify-between. Map container uses h-full w-full. No fixed pixel widths on primary layout containers except Sidebar. Functional at 1280px+. Mobile not in scope for Episode format.	✓ PASS
D7	Functional Sidebar 30% Width	Sidebar must be exactly 30% of viewport	Sidebar wrapper updated to style={{width:'30%'}} with flex-none. DNA-compliant at all resolutions. Fix verified in page.tsx.	✓ PASS
E1	Intelligence Value Economic Context	Dashboard provides meaningful insight, not just data display	Sidebar Section B explains: G20 5% cost target, average corridor cost vs target gap, migrant worker economic impact. Section C maps regulatory actors. Intelligence framing is strong.	✓ PASS
E2	Intelligence Value Why This Matters Copy	Explains to everyday viewers, builders, and allocators	Copy in Sidebar.tsx uses plain English. Avoids jargon. Explains remittance cost as a % of \$200 send — immediately relatable. Passes the 'everyday viewer' test.	✓ PASS
E3	Intelligence Value Source Attribution	Data sources attributed with refresh context	Sidebar footer shows: 'World Bank Remittance Prices · ECB FX Rates · Synthetic Unit Economics'. Data source badge in header shows LIVE/MOCK status. Source attribution present.	✓ PASS
F1	New Feature Corridor Heatmap	Map arcs colour-coded by volume intensity	heat_intensity added to all 15 corridors in mock_data.json. /api/heatmap returns heat_color per corridor. CorridorMap.tsx colours arcs: red (very high), amber (high), cyan (medium), indigo (low). Heatmap legend visible on map. Corridor tabs reflect heat colour.	✓ PASS
F2	New Feature Cost vs Income Overlay	Scatter chart: fee % vs sender income level	sender_avg_income_usd and fee_pct_of_income added to all 15 corridors. /api/cost-income-overlay returns burden_level (CRITICAL/HIGH/MEDIUM/LOW). CostIncomeChart.tsx renders scatter chart with G20 5% reference line. Main stage tab.	✓ PASS

#	AUDIT CATEGORY	CHECK	FINDING	VERDICT
F3	New Feature Player Breakdown	Fintech vs Traditional MTO vs Bank comparison	player_type added to all 46 providers. /api/player-breakdown groups by type, computes avg fee + FX margin. PlayerBreakdownChart.tsx renders bar chart. Available in sidebar Players tab.	✓ PASS
F4	New Feature FX Margin Visualizer	Hidden spread vs advertised fee stacked bar	fx_margin added to all providers (Wise: 0.4%, WU: 2.8%). /api/fx-margin returns advertised_fee + hidden_spread + true_cost. FXMarginChart.tsx stacked bar: green (advertised) + amber (hidden). Sidebar FX Margin tab.	✓ PASS
F5	New Feature Volume Flow Tracker	All 15 corridors ranked by total volume	/api/volume-flow returns corridors sorted by volume_bn_usd with volume_share_pct. page.tsx renders ranked table as main stage tab. Clickable rows select corridor. Total \$240B tracked. Colour-coded by G20 compliance.	✓ PASS

TOTAL CHECKS	PASS	IMPROVE	FAIL	SCORE
33	33	0	0	33/33

✓ VAR STATUS: GREEN · ALL CHECKS PASSED · v4.0 FEATURE EXPANSION VERIFIED

The following items are rated IMPROVE (non-blocking). Apply these fixes before client handoff.

B2 / D7

Sidebar 30% Width Fix

Replace w-[340px] with a percentage-based flex value so the 70/30 split holds at all resolutions.

```
// In page.tsx – replace the Sidebar wrapper: // BEFORE: // AFTER:
```